

Ruprecht-Karls-Universität Heidelberg
Fakultät für Ingenieurwissenschaften
Studiengang Molekulare Biotechnologie

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Declaration of Independence

I hereby declare on my honor that:

1. I have written this Bachelor thesis independently under supervision and have used no sources or aids other than those indicated;
2. any literal quotations from literature/internet as well as the use of ideas from other authors have been identified as such at the appropriate places within the thesis;
3. I have not submitted my Bachelor thesis for any other examination.

I am aware that a false declaration will have legal consequences.

Place, Date

Signature

Note:

Without signature, date, and place, this declaration is invalid and so is the Bachelor thesis.

Acknowledgements

Here, you can thank all your contributors.

Abstract

Your Abstract goes here.

Zusammenfassung

Hier eine deutsche Übersetzung des Abstract.

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Lists and Abbreviations

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List of Abbreviations

abbr	long form	4
VSCode	Visual Studio Code	V
IDE	integrated development environment	1
APC	antigen-presenting cell	2

1 Background

Welcome to this LaTeX template for the Bachelor's in MoBi. This template was made to adhere to the standards of 2024, and I cannot guarantee that it will work as intended or be exactly what the institute wants from you.

1.1 Intro to LaTeX

LaTeX is a language for writing and formatting serious documents in a reproducible way. Unlike Microsoft Word, you will not see the text appear in the document directly as you write it. Instead, you have to run the code in order to "build" your output PDF.

1.1.1 Installing LaTeX

As with any other "programming language", you have to install it before you can use it. Depending on your operating system it might already be installed, but you can also get it from [here](#).

1.1.2 LaTeX in VSCode

VSCode is a widely used integrated development environment (IDE). It can be extended by installing extensions from community members or organizations. For writing LaTeX code you can use the LaTeX Workshop extension by James-Yu.

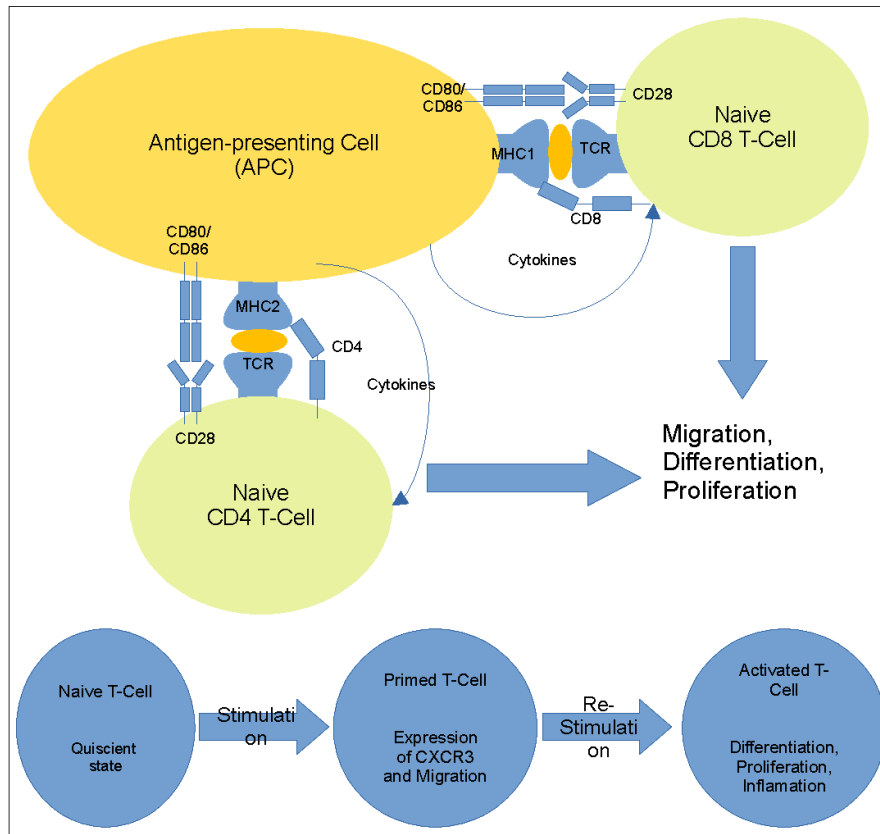


Figure 1: T-Cell Activation.

T-cells are activated by antigen-presenting cells (APCs), by the interaction of TCR with MHC and Antigen and various cofactors. The specific mix of cofactors decides the subtype of activated CD4 T-cells. After activation in the lymph node, T-cells migrate through the circulatory system until they find a locus of inflammation, where they are restimulated by another APC carrying the same antigen.

2 Methods

2.1 Line wrap in VSCode

To auto-wrap lines after a certain number of characters, go to settings by pressing F1 and typing ">Preferences: Open Settings (UI)". In the top-level tabs, go to Workspace and search for the Word Wrap setting. I suggest setting it to "bounded".

2.2 Figures

To insert a figure, you can use the premade code block below. For more customization, you can look up overleaf.com. Figures will not necessarily appear where you placed them in the code; but they appear where LaTeX thinks they fit best. You can modify this with tags after `\begin{figure}`.

2.2.1 Figure Format

For good image resolution, always save your figures/plots as vectorized files, i.e. PDFs or SVGs. Since inserting a lot of PDFs into a LaTeX file makes the load time much longer, converting them to PNG and inserting the PNGs instead is much more efficient during writing. For this, you can use the `convert_figures.sh` bash script, which creates converted copies of all PDFs in the `figures` and `images` directories. To tell LaTeX to use the PNGs instead of the PDFs, uncomment the `\DeclareGraphicsExtensions{.png,.pdf}` line in the `main/BachelorsIPMB.sty` file. Then, when you want to build the final version, comment the line again to use the PDF versions.

2.3 Some tables

Here is an example of how to use a CSV file as a table. There are other ways to create tables, such as this one. Another way to insert a table is also used in the AI tools section in the supplement.

Table 1: Baseline Hyperparameters.

Parameter	Value
genes	250
genes per context for training	20
genes per context for testing	5,000
training epochs	1,000
batch size	0.2
validation size	$1 \cdot 10^{-2}$
learning rate	0.1

2.4 Errors

At some point you will encounter an error message popping up when you are building your document. In VSCode, handling this is not straightforward. You can look back at your last edit and see what might be causing it, or you can search the log file for error messages.

2.5 Citations

To import your citations, add your `".bib"` file to the `misc` directory and update the `addbibresource` line in `main/main.tex` to match your file name. You can then insert citations using `[1]`. The bibliography will be automatically created at the end of the document by `printbibliography`.

2.6 Referencing Figures, Tables and Sections

To refer to figures or sections by their numbers, label the element with `label` (as in this section). Then you can reference its number by using `ref`. For this chapter you could do: see section 2.6.

Or you can reference figure 1 or table 1.

2.7 Abbreviations

To automate abbreviations, this document uses the acronym package. New abbreviations are defined in main.tex using `\acro{label}[abbr]{long form}`. To reference them in the text, use `\ac`. The first occurrence will be the long form with the abbreviation in brackets. There are several options though:

- To skip counting the first occurrence, as in the abstract, use `\ac*{label}: long form (abbr)`
- To capitalize the long form use `\Ac{label}: Long form (abbr)`
- To use in basic mode `\ac{label}: long form (abbr)`
- To insert the plural use `\acp{label}: abbrs`
- To force the long form, use `\acf{label}: long form (abbr)`
- Finally, you can combine any of these options like: `\Acp*{label}: Long forms (abbrs)`

For even more options, you can look up the package's documentation on ctan.

2.8 List of Figures, Tables and Abbreviations

The lists of figures, tables and abbreviations will be automatically updated as you add more of those elements to the document.

2.9 Tips for self-help

To find specific tips and tutorials, you can search the internet, and in most cases you will find a helpful blog post, forum thread or Overleaf tutorial.

For automating figure insertion, method drafts and other verifiable tasks, I recommend using an agentic LLM system such as Claude Code.

3 Results

Your results go here.

4 Discussion

Your discussion goes here.

References

1. Abramson, J. *et al.* Accurate Structure Prediction of Biomolecular Interactions with AlphaFold 3. *Nature* **630**, 493–500. ISSN: 1476-4687. (2026) (June 2024).

A Supplement

Your supplementary data and figures go here.

A.1 AI Tools

Table 2: List of AI Tools used in this work.

Tool	Goal	Part of the thesis	Notes
DeepL	Translation and Synonyms	Introduction and Methods	DeepL's standard translator was used to translate german and english.
GitHub Copilot	Coding Support	Figures	Copilot was used to assist in plotting.